Part III Cheeger – Colding Theory

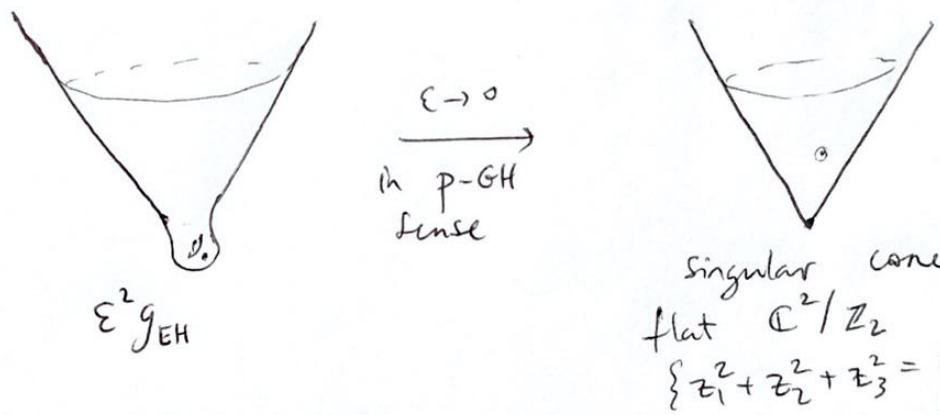
E.g. Eguchi – Hanson metric g_{EH} on

$$M = \{ z_1^2 + z_2^2 + z_3^2 = 1 \} \subset \mathbb{C}^3$$

$Ric(g_{EH}) = 0$; $\exists\, v > 0$ s.t.

$$\mathrm{Vol}\, B(p, r) \geq v r^4 \quad \text{for all } r > 0$$

g_{EH} is asymptotic to flat cone $\mathbb{C}^2/\mathbb{Z}_2$



Cheeger – Colding theory:

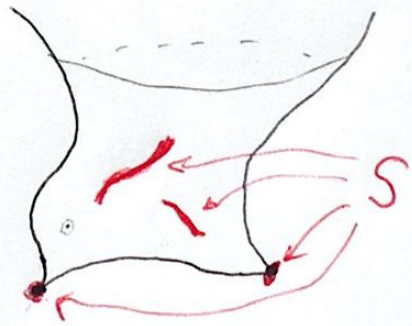
Similar picture holds in general:

If (M_i, g_i) satisfy $Ric(g_i) = \lambda_i g_i$, $|\lambda_i| \leq 1$

and $\mathrm{Vol}\, B(p, 1) \geq v > 0$

then subsequence has pointed GH-limit Z

Z is smooth away from codim 4 singular set S and convergence to Z is in $C^{k,\alpha}$ on compact sets away from S



Z is modeled on cones near points of S.

<u>More precise overview</u>

Let (M_i, g_i, p_i) be n-folds s.t.

- $\mathrm{Ric}(g_i) > -g_i$
- $\mathrm{Vol}\, B(p_i, 1) > v > 0$ (non-collapsing)
- $(M_i, g_i, p_i) \longrightarrow (Z, d_Z, p)$ in pointed GH sense.

$\forall q \in Z$ and $a_k \to \infty$ consider sequence

$$(Z, a_k \cdot d_Z, q)$$

up to choosing subsequence get limit Z_q called a tangent cone to Z at q.

[<u>Remark</u>: Z_q is also obtained as pointed limit of $(M_{i_k}, a_k^2 \cdot g_{i_k}, q_{i_k})$ where $q_{i_k} \to q$]

<u>Thm</u> Each Z_q is a metric cone

Define stratification

$$S_0 \subset S_1 \subset \dots \subset S_{n-1} \subset Z$$

$$S_k = \left\{ q \in Z \;\middle|\; \begin{array}{l} \text{no tangent cone } Z_q \text{ at } q \\ \text{is of form } Z_q = \mathbb{R}^{k+1} \times Y \end{array} \right\}$$

E.g. $q \in Z \setminus S_{n-1} \iff$ some tangent cone $Z_q = \mathbb{R}^n$

In this case all tangent cones at q are $\mathbb{R}^n$.

$Z \setminus S_{n-1} = \mathcal{R}$ is the regular set

Thm
- $\dim_{\mathcal{H}} S_k \leq k$
- $S_{n-1} = S_{n-2} \implies \dim(Z \setminus \mathcal{R}) \leq n-2$

If also $|Ric(g_i)| \leq 1$ along original sequence, then
- (Cheeger–Naber) $\dim_{\mathcal{H}} Z \setminus \mathcal{R} \leq n-4$
- (Anderson) $\mathcal{R}$ is open smooth Riem mfld $(\mathcal{R}, g)$ and $\forall\, U \subset\subset \mathcal{R}$ $\exists$ embeddings $\varphi_i : U \hookrightarrow M_i$ for large i s.t. $\varphi_i^* g_i \longrightarrow g$ in $C^{1,\alpha}$

Ideas for proving that tangent cones are metric cones:

Riemannian cones (C, g)

$$g = dr^2 + r^2 g_L \qquad (L, g_L) = \text{link of cone}$$

are characterized by the property

$$\mathrm{Hess}_{r^2} = 2g$$

Main points:

(1) If we find a function f on $B = B(p,1)$ s.t. $\| \mathrm{Hess}_f - 2g \|_{L^2(B)}$ is small then B is GH-close to ball in metric cone.

(2) If $\mathrm{Ric} \geq 0$ and

$$\mathrm{Vol}\, B(p,2) \geq 2^n \,\mathrm{Vol}\, B(p,1) \cdot (1-\delta)$$

$\Rightarrow$ using Bochner formula construct f as in (1):

$$\Delta f = 2n \quad \text{in } B(p,1)$$
$$f = 1 \quad \text{on } \partial B(p,1)$$

(3) Let $\mathrm{Vol}\, B(p,1) \geq v > 0$, $\mathrm{Ric} \geq 0$

Bishop-Gromov implies

$$\frac{\mathrm{Vol}\, B(p,(\frac{1}{2})^k)}{(\frac{1}{2})^{kn}\,\omega_n} \quad \text{increases with } k$$

$$\frac{\mathrm{Vol}\, B(p,1)}{\omega_n} \geq \frac{v}{\omega_n} \quad \text{as } k \to 0$$

$$1 \quad \text{as } k \to \infty$$

Given any $\delta > 0$ there are $\leq N(\delta, v)$ integers k s.t.

$$2^n \,\mathrm{Vol}\, B(p,(\tfrac{1}{2})^{k+1}) \geq \mathrm{Vol}\, B(p,(\tfrac{1}{2})^k) \cdot (1+\delta)$$

For all remaining k can apply (2) & (1) to see $B(p,(\frac{1}{2})^{k+1})$ is GH close to a cone